A Family and selection-rule comparisons

All construct panels use the same 305 recordings and two-rater references: model agreement averages comparisons against the three possible two-human means, while human consensus agreement compares each human with the other two. Each family occupies one row, with selection conditions arranged horizontally; human and Azure benchmarks appear above the family comparisons because selection rules do not apply to those benchmarks. Marginal 95% intervals use 2,000 speaker-bootstrap draws (seed 42). Complete machine-readable results, including three-rater comparisons, remain in the aggregate release.

Three public-development selection conditions were fixed before this comparison: S selects each construct by Spearman on $\{0.001, 0.01, 0.1, 1, 10, 100\}$; D selects by development accuracy MAE on the same grid; H selects by accuracy MAE on $\{0.00005, 0.0005, 0.005, 0.05, 0.5, 5\}$. The historical summed-loss penalties $\{0.1, 1, 10, 100, 1000, 10000\}$ were divided by the original 2,000 SpeechOcean training utterances to express the same penalty strength under normalized loss. That fixed conversion is applied to every family, not divided again by its sample count. Corpus-balanced weights, training-only scaling, feature combinations, and partitions are identical across conditions. Development metrics are averaged over the included corpora. S breaks ties by lower MAE, then larger penalty; D and H break ties by larger penalty. Accuracy MAE selects one penalty per representation combination, so fluency and prosody share the penalty selected for their common XLS-R + XEUS features; their output coefficients remain separate. H reproduces the historical grid within the matched fitting convention, not every historical implementation detail. All 21 family–condition combinations completed all 305 recordings. The main table displays the external leaders across S, D, and H, including every configuration within .001 of the highest unrounded correlation or QWK for each construct. The three construct panels below preserve the common-condition comparisons; the matched SpeechOcean refit replaces the historical deployment in S. No private ratings entered coefficient estimation, tuning, or calibration. Choosing a deployment from these external comparisons would be a post-evaluation choice. The separate historical deployed SpeechOcean scores remain in the released aggregate file and are not substituted for the matched S row. Historical-grid DAMAE improves koreanocean two-rater fluency QWK from 0.585 to 0.744 and prosody QWK from 0.546 to 0.675, while the corresponding correlations change from 0.840 to 0.838 and 0.850 to 0.840. The fluency QWK point estimate is slightly above the human baseline of 0.739; the paired interval for that difference includes zero. Different rules therefore trade rank association against numerical agreement, without establishing one uniformly best condition. The intervals below are marginal, not intervals for condition differences; the 258-contrast adjustment in the following section covers all three conditions. The released aggregate CSV preserves both two- and three-rater comparisons and their intervals, along with the human pairwise baseline.

Table 1: All families and selection rules by construct ($N = 305$).

Accuracy

Benchmark	Spearman ρ [95% CI]	QWK [95% CI]
Human consensus	**0.809** [0.769,0.842]	**0.704** [0.663,0.743]
Azure	0.761 [0.709,0.804]	0.161 [0.129,0.196]

Family	S: Spearman		D: Accuracy MAE		H: Historical-grid MAE	
	ρ	QWK	ρ	QWK	ρ	QWK
	0.856	0.502	0.856	0.502	0.844	0.546
speechocean	[0.821,0.882]	[0.451,0.549]	[0.821,0.882]	[0.451,0.549]	[0.807,0.872]	[0.494,0.592]
	0.856	0.702	0.856	0.702	0.853	0.672
korean	[0.822,0.882]	[0.662,0.739]	[0.822,0.882]	[0.662,0.739]	[0.820,0.881]	[0.631,0.711]
	0.852	0.241	0.864	0.239	0.863	0.239
aesop	[0.818,0.878]	[0.215,0.265]	[0.832,0.888]	[0.214,0.264]	[0.830,0.887]	[0.214,0.264]
	0.853	**0.821**	0.841	0.798	0.849	0.819
koreanocean	[0.816,0.882]	[0.789,0.847]	[0.802,0.871]	[0.764,0.827]	[0.811,0.879]	[0.788,0.846]
	0.872	0.598	0.865	0.628	**0.872**	0.607
aesopocean	[0.842,0.894]	[0.563,0.631]	[0.833,0.890]	[0.593,0.662]	[0.841,0.894]	[0.572,0.640]
	0.867	0.477	0.867	0.477	0.867	0.472
koreanaesop	[0.835,0.891]	[0.439,0.512]	[0.835,0.891]	[0.439,0.512]	[0.836,0.891]	[0.435,0.508]
	0.870	0.585	0.862	0.576	0.857	0.575
aesopkoreanocean	[0.839,0.894]	[0.548,0.620]	[0.829,0.888]	[0.537,0.613]	[0.824,0.884]	[0.537,0.611]

Fluency

Benchmark	Spearman ρ [95% CI]	QWK [95% CI]
Human consensus	**0.807** [0.768,0.840]	**0.739** [0.695,0.777]
Azure	0.633 [0.556,0.698]	0.078 [0.056,0.107]

Family	S: Spearman		D: Accuracy MAE		H: Historical-grid MAE	
	ρ	QWK	ρ	QWK	ρ	QWK
	0.811	0.716	0.847	0.629	0.842	0.670
speechocean	[0.763,0.848]	[0.672,0.752]	[0.806,0.878]	[0.585,0.666]	[0.801,0.874]	[0.627,0.706]
	0.860	0.495	0.860	0.495	0.859	0.501
korean	[0.825,0.888]	[0.452,0.535]	[0.825,0.888]	[0.452,0.535]	[0.825,0.886]	[0.458,0.538]
	0.850	0.276	0.855	0.312	0.859	0.315
aesop	[0.813,0.879]	[0.247,0.305]	[0.819,0.883]	[0.279,0.342]	[0.823,0.885]	[0.282,0.346]
	0.840	0.585	0.846	0.714	0.838	**0.744**
koreanocean	[0.799,0.873]	[0.537,0.628]	[0.808,0.877]	[0.672,0.751]	[0.799,0.870]	[0.702,0.779]
	0.874	0.613	0.874	0.613	0.862	0.612
aesopocean	[0.842,0.899]	[0.575,0.647]	[0.842,0.899]	[0.575,0.647]	[0.828,0.890]	[0.574,0.647]
	0.876	0.462	0.876	0.462	**0.878**	0.469
koreanaesop	[0.846,0.900]	[0.424,0.496]	[0.846,0.900]	[0.424,0.496]	[0.848,0.901]	[0.431,0.502]
	0.874	0.527	**0.877**	0.577	0.871	0.591
aesopkoreanocean	[0.842,0.898]	[0.486,0.564]	[0.846,0.902]	[0.539,0.612]	[0.839,0.897]	[0.552,0.626]

Prosody

Benchmark	Spearman ρ [95% CI]	QWK [95% CI]
Human consensus	**0.807** [0.768,0.842]	**0.742** [0.701,0.778]
Azure	0.737 [0.672,0.792]	0.370 [0.318,0.421]

Family	S: Spearman		D: Accuracy MAE		H: Historical-grid MAE	
	ρ	QWK	ρ	QWK	ρ	QWK
	0.819	0.597	0.850	0.545	0.847	0.570
speechocean	[0.776,0.855]	[0.551,0.639]	[0.813,0.880]	[0.501,0.585]	[0.809,0.878]	[0.525,0.610]
	0.874	0.489	0.874	0.489	0.872	0.493
korean	[0.842,0.899]	[0.447,0.526]	[0.842,0.899]	[0.447,0.526]	[0.840,0.896]	[0.452,0.531]
	0.850	0.261	0.855	0.280	0.858	0.283
aesop	[0.814,0.878]	[0.232,0.288]	[0.820,0.882]	[0.249,0.309]	[0.824,0.884]	[0.252,0.312]
	0.850	0.546	0.851	0.641	0.840	**0.675**
koreanocean	[0.812,0.881]	[0.501,0.586]	[0.815,0.882]	[0.599,0.680]	[0.802,0.872]	[0.632,0.712]
	0.881	0.541	0.870	0.497	0.861	0.481
aesopocean	[0.850,0.905]	[0.502,0.576]	[0.838,0.895]	[0.457,0.531]	[0.827,0.888]	[0.440,0.516]
	0.883	0.431	**0.883**	0.431	**0.883**	0.433
koreanaesop	[0.852,0.906]	[0.394,0.465]	[0.852,0.906]	[0.394,0.465]	[0.853,0.906]	[0.397,0.468]
	0.881	0.480	0.879	0.505	0.872	0.518
aesopkoreanocean	[0.850,0.906]	[0.440,0.516]	[0.847,0.904]	[0.467,0.541]	[0.839,0.898]	[0.479,0.554]

Note. Cells report estimates and marginal 95% intervals. Bold marks the human benchmark and model estimates within .001 of each construct–metric maximum across 21 family–rule combinations. Near-ties are descriptive. Accuracy clipping: AESOP has one below-zero prediction under S and two under each of D and H; none exceeds 100. No fluency or prosody prediction requires clipping. QWK caps out-of-range predictions at 0 or 100, including bootstrap intervals; Spearman retains raw predictions. Paired tests cover S/D/H.

B Paired differences and score distributions

The same 305 recordings were jointly resampled 600,000 times per seed (42 and 2026), treating each unique recording URL as one speaker. Each replicate recomputed agreement against the same three two-rater reference pairs. Bonferroni intervals use percentile endpoints $0.025/258$ and $1 - 0.025/258$, covering 21 models versus human consensus and Azure, plus Azure versus human consensus, for three constructs and two metrics. Both seeds agree on all 258 adjusted interval-exclusion decisions. The table reports seed-42 intervals; full-precision results for both seeds are released. Approximately 58 draws fall in each adjusted tail, so endpoints remain Monte Carlo approximations. Correction covers these contrasts, not the preceding model search or direct comparisons between selection rules; inference conditions on the observed panel and fitted models. No private scores were used for fitting or calibration.

Table 2: Paired differences for all S/D/H configurations and constructs.

			Accuracy			Fluency			Prosody		
System	Reference	Metric	Diff.	95% CI	Adjusted CI	Diff.	95% CI	Adjusted CI	Diff.	95% CI	Adjusted CI
Azure benchmark (independent of selection rule)											
azure	Human	ρ	-0.049	[-0.096, -0.005]	[-0.143, 0.036]	-0.174	[-0.243, -0.110]	[-0.315, -0.057]	-0.070	[-0.125, -0.022]	[-0.183, 0.016]
azure	Human	QWK	-0.543	[-0.580, -0.504]	[-0.610, -0.466]	-0.661	[-0.697, -0.621]	[-0.727, -0.581]	-0.372	[-0.419, -0.325]	[-0.461, -0.282]
S: construct-specific Spearman											
speechocean	Human	ρ	+0.046	[0.016, 0.076]	[-0.013, 0.106]	+0.004	[-0.036, 0.041]	[-0.076, 0.075]	+0.012	[-0.023, 0.044]	[-0.058, 0.074]
speechocean	Human	QWK	-0.202	[-0.249, -0.155]	[-0.290, -0.111]	-0.023	[-0.067, 0.022]	[-0.109, 0.062]	-0.144	[-0.188, -0.101]	[-0.226, -0.060]
speechocean	Azure	ρ	+0.095	[0.054, 0.139]	[0.017, 0.185]	+0.178	[0.113, 0.247]	[0.056, 0.314]	+0.082	[0.033, 0.135]	[-0.007, 0.193]
speechocean	Azure	QWK	+0.341	[0.299, 0.381]	[0.262, 0.416]	+0.638	[0.601, 0.671]	[0.565, 0.700]	+0.228	[0.173, 0.282]	[0.122, 0.329]
korean	Human	ρ	+0.046	[0.018, 0.075]	[-0.009, 0.106]	+0.054	[0.022, 0.085]	[-0.007, 0.117]	+0.067	[0.039, 0.095]	[0.014, 0.125]
korean	Human	QWK	-0.002	[-0.041, 0.037]	[-0.077, 0.073]	-0.244	[-0.281, -0.208]	[-0.313, -0.175]	-0.253	[-0.287, -0.219]	[-0.317, -0.189]
korean	Azure	ρ	+0.095	[0.055, 0.139]	[0.021, 0.184]	+0.228	[0.164, 0.297]	[0.111, 0.366]	+0.137	[0.098, 0.183]	[0.069, 0.233]
korean	Azure	QWK	+0.541	[0.500, 0.580]	[0.464, 0.614]	+0.417	[0.379, 0.452]	[0.346, 0.483]	+0.119	[0.077, 0.160]	[0.037, 0.195]
aesop	Human	ρ	+0.043	[0.011, 0.074]	[-0.017, 0.106]	+0.044	[0.009, 0.077]	[-0.022, 0.110]	+0.043	[0.012, 0.073]	[-0.017, 0.103]
aesop	Human	QWK	-0.463	[-0.493, -0.431]	[-0.519, -0.400]	-0.463	[-0.494, -0.430]	[-0.521, -0.398]	-0.481	[-0.510, -0.451]	[-0.534, -0.422]
aesop	Azure	ρ	+0.092	[0.054, 0.132]	[0.022, 0.174]	+0.218	[0.157, 0.284]	[0.106, 0.351]	+0.113	[0.073, 0.159]	[0.043, 0.210]
aesop	Azure	QWK	+0.080	[0.053, 0.105]	[0.026, 0.126]	+0.198	[0.174, 0.221]	[0.151, 0.241]	-0.109	[-0.147, -0.072]	[-0.182, -0.040]
koreanocean	Human	ρ	+0.044	[0.013, 0.074]	[-0.018, 0.105]	+0.033	[-0.004, 0.068]	[-0.041, 0.102]	+0.042	[0.010, 0.074]	[-0.023, 0.103]
koreanocean	Human	QWK	+0.116	[0.082, 0.152]	[0.051, 0.187]	-0.154	[-0.195, -0.114]	[-0.232, -0.078]	-0.196	[-0.232, -0.161]	[-0.265, -0.129]
koreanocean	Azure	ρ	+0.093	[0.051, 0.137]	[0.015, 0.183]	+0.207	[0.140, 0.279]	[0.084, 0.351]	+0.113	[0.069, 0.163]	[0.034, 0.217]
koreanocean	Azure	QWK	+0.660	[0.626, 0.690]	[0.595, 0.715]	+0.507	[0.464, 0.546]	[0.423, 0.580]	+0.176	[0.128, 0.223]	[0.081, 0.265]
aesopocean	Human	ρ	+0.062	[0.033, 0.093]	[0.006, 0.124]	+0.068	[0.038, 0.098]	[0.011, 0.129]	+0.073	[0.048, 0.100]	[0.025, 0.127]
aesopocean	Human	QWK	-0.106	[-0.140, -0.071]	[-0.170, -0.039]	-0.126	[-0.159, -0.093]	[-0.189, -0.062]	-0.201	[-0.232, -0.170]	[-0.260, -0.141]
aesopocean	Azure	ρ	+0.111	[0.071, 0.155]	[0.037, 0.202]	+0.242	[0.180, 0.309]	[0.129, 0.379]	+0.144	[0.099, 0.195]	[0.066, 0.251]
aesopocean	Azure	QWK	+0.437	[0.402, 0.470]	[0.370, 0.500]	+0.535	[0.501, 0.566]	[0.469, 0.593]	+0.171	[0.128, 0.214]	[0.087, 0.251]
koreanaesop	Human	ρ	+0.057	[0.028, 0.088]	[0.000, 0.119]	+0.069	[0.040, 0.099]	[0.014, 0.130]	+0.076	[0.049, 0.104]	[0.026, 0.133]
koreanaesop	Human	QWK	-0.227	[-0.261, -0.193]	[-0.290, -0.161]	-0.277	[-0.311, -0.243]	[-0.340, -0.212]	-0.311	[-0.342, -0.279]	[-0.370, -0.250]
koreanaesop	Azure	ρ	+0.106	[0.066, 0.150]	[0.032, 0.196]	+0.243	[0.182, 0.311]	[0.132, 0.379]	+0.146	[0.105, 0.194]	[0.075, 0.246]
koreanaesop	Azure	QWK	+0.316	[0.281, 0.350]	[0.249, 0.380]	+0.384	[0.350, 0.416]	[0.320, 0.444]	+0.062	[0.022, 0.100]	[-0.015, 0.133]
aesopkoreanocean	Human	ρ	+0.061	[0.032, 0.091]	[0.005, 0.123]	+0.067	[0.037, 0.098]	[0.010, 0.128]	+0.074	[0.047, 0.101]	[0.022, 0.129]
aesopkoreanocean	Human	QWK	-0.119	[-0.153, -0.084]	[-0.183, -0.051]	-0.212	[-0.247, -0.178]	[-0.278, -0.148]	-0.262	[-0.293, -0.231]	[-0.320, -0.203]
aesopkoreanocean	Azure	ρ	+0.110	[0.070, 0.154]	[0.037, 0.200]	+0.241	[0.178, 0.311]	[0.127, 0.382]	+0.144	[0.101, 0.194]	[0.069, 0.249]
aesopkoreanocean	Azure	QWK	+0.425	[0.388, 0.459]	[0.356, 0.490]	+0.449	[0.412, 0.482]	[0.378, 0.512]	+0.111	[0.068, 0.151]	[0.029, 0.187]
D: accuracy MAE											
speechocean	Human	ρ	+0.046	[0.016, 0.076]	[-0.013, 0.106]	+0.040	[0.006, 0.073]	[-0.029, 0.104]	+0.043	[0.012, 0.072]	[-0.018, 0.101]
speechocean	Human	QWK	-0.202	[-0.249, -0.155]	[-0.290, -0.111]	-0.111	[-0.156, -0.064]	[-0.195, -0.020]	-0.196	[-0.240, -0.151]	[-0.278, -0.109]
speechocean	Azure	ρ	+0.095	[0.054, 0.139]	[0.017, 0.185]	+0.214	[0.150, 0.283]	[0.097, 0.353]	+0.113	[0.066, 0.166]	[0.029, 0.224]
speechocean	Azure	QWK	+0.341	[0.299, 0.381]	[0.262, 0.416]	+0.550	[0.511, 0.586]	[0.474, 0.617]	+0.176	[0.119, 0.232]	[0.067, 0.282]
korean	Human	ρ	+0.046	[0.018, 0.075]	[-0.009, 0.106]	+0.054	[0.022, 0.085]	[-0.007, 0.117]	+0.067	[0.039, 0.095]	[0.014, 0.125]
korean	Human	QWK	-0.002	[-0.041, 0.037]	[-0.077, 0.073]	-0.244	[-0.281, -0.208]	[-0.313, -0.175]	-0.253	[-0.287, -0.219]	[-0.317, -0.189]
korean	Azure	ρ	+0.095	[0.055, 0.139]	[0.021, 0.184]	+0.228	[0.164, 0.297]	[0.111, 0.366]	+0.137	[0.098, 0.183]	[0.069, 0.233]
korean	Azure	QWK	+0.541	[0.500, 0.580]	[0.464, 0.614]	+0.417	[0.379, 0.452]	[0.346, 0.483]	+0.119	[0.077, 0.160]	[0.037, 0.195]

			Accuracy			Fluency			Prosody		
System	Reference	Metric	Diff.	95% CI	Adjusted CI	Diff.	95% CI	Adjusted CI	Diff.	95% CI	Adjusted CI
aesop	Human	ρ	+0.054	[0.025, 0.085]	[-0.003, 0.117]	+0.048	[0.015, 0.081]	[-0.016, 0.114]	+0.048	[0.018, 0.078]	[-0.010, 0.107]
aesop	Human	QWK	-0.465	[-0.493, -0.434]	[-0.519, -0.403]	-0.428	[-0.459, -0.395]	[-0.486, -0.365]	-0.462	[-0.491, -0.432]	[-0.516, -0.403]
aesop	Azure	ρ	+0.103	[0.064, 0.146]	[0.032, 0.190]	+0.222	[0.163, 0.288]	[0.113, 0.355]	+0.119	[0.077, 0.166]	[0.045, 0.216]
aesop	Azure	QWK	+0.079	[0.052, 0.103]	[0.026, 0.124]	+0.233	[0.206, 0.259]	[0.182, 0.283]	-0.090	[-0.128, -0.053]	[-0.162, -0.021]
koreanocean	Human	ρ	+0.031	[-0.000, 0.062]	[-0.032, 0.092]	+0.039	[0.005, 0.072]	[-0.028, 0.103]	+0.044	[0.014, 0.073]	[-0.015, 0.102]
koreanocean	Human	QWK	+0.093	[0.058, 0.129]	[0.027, 0.163]	-0.025	[-0.063, 0.012]	[-0.098, 0.046]	-0.101	[-0.135, -0.067]	[-0.165, -0.037]
koreanocean	Azure	ρ	+0.080	[0.037, 0.125]	[-0.000, 0.171]	+0.213	[0.148, 0.284]	[0.094, 0.354]	+0.114	[0.071, 0.164]	[0.039, 0.217]
koreanocean	Azure	QWK	+0.637	[0.600, 0.671]	[0.567, 0.698]	+0.636	[0.595, 0.672]	[0.557, 0.703]	+0.272	[0.225, 0.317]	[0.181, 0.357]
aesopocean	Human	ρ	+0.056	[0.025, 0.086]	[-0.002, 0.118]	+0.068	[0.038, 0.098]	[0.011, 0.129]	+0.063	[0.037, 0.089]	[0.013, 0.116]
aesopocean	Human	QWK	-0.076	[-0.109, -0.041]	[-0.139, -0.009]	-0.126	[-0.159, -0.093]	[-0.189, -0.062]	-0.245	[-0.276, -0.214]	[-0.304, -0.185]
aesopocean	Azure	ρ	+0.105	[0.063, 0.150]	[0.027, 0.198]	+0.242	[0.180, 0.309]	[0.129, 0.379]	+0.133	[0.089, 0.184]	[0.055, 0.240]
aesopocean	Azure	QWK	+0.468	[0.431, 0.502]	[0.399, 0.531]	+0.535	[0.501, 0.566]	[0.469, 0.593]	+0.127	[0.085, 0.168]	[0.046, 0.205]
koreanaesop	Human	ρ	+0.057	[0.028, 0.088]	[0.000, 0.119]	+0.069	[0.040, 0.099]	[0.014, 0.130]	+0.076	[0.049, 0.104]	[0.026, 0.133]
koreanaesop	Human	QWK	-0.227	[-0.261, -0.193]	[-0.290, -0.161]	-0.277	[-0.311, -0.243]	[-0.340, -0.212]	-0.311	[-0.342, -0.279]	[-0.370, -0.250]
koreanaesop	Azure	ρ	+0.106	[0.066, 0.150]	[0.032, 0.196]	+0.243	[0.182, 0.311]	[0.132, 0.379]	+0.146	[0.105, 0.194]	[0.075, 0.246]
koreanaesop	Azure	QWK	+0.316	[0.281, 0.350]	[0.249, 0.380]	+0.384	[0.350, 0.416]	[0.320, 0.444]	+0.062	[0.022, 0.100]	[-0.015, 0.133]
aesopkoreanocean	Human	ρ	+0.052	[0.022, 0.083]	[-0.006, 0.114]	+0.070	[0.040, 0.101]	[0.013, 0.132]	+0.072	[0.045, 0.099]	[0.020, 0.127]
aesopkoreanocean	Human	QWK	-0.128	[-0.162, -0.094]	[-0.193, -0.062]	-0.162	[-0.196, -0.128]	[-0.227, -0.097]	-0.237	[-0.268, -0.205]	[-0.296, -0.176]
aesopkoreanocean	Azure	ρ	+0.101	[0.060, 0.146]	[0.026, 0.191]	+0.244	[0.182, 0.313]	[0.130, 0.384]	+0.142	[0.099, 0.191]	[0.069, 0.245]
aesopkoreanocean	Azure	QWK	+0.415	[0.378, 0.451]	[0.344, 0.482]	+0.499	[0.463, 0.532]	[0.430, 0.561]	+0.136	[0.095, 0.176]	[0.056, 0.211]
H: historical-grid accuracy MAE											
speechocean	Human	ρ	+0.035	[0.004, 0.065]	[-0.027, 0.095]	+0.035	[-0.000, 0.069]	[-0.036, 0.100]	+0.040	[0.009, 0.069]	[-0.022, 0.097]
speechocean	Human	QWK	-0.158	[-0.206, -0.110]	[-0.248, -0.066]	-0.069	[-0.113, -0.024]	[-0.152, 0.018]	-0.172	[-0.215, -0.128]	[-0.251, -0.088]
speechocean	Azure	ρ	+0.083	[0.041, 0.128]	[0.004, 0.174]	+0.209	[0.145, 0.278]	[0.090, 0.348]	+0.110	[0.063, 0.163]	[0.026, 0.220]
speechocean	Azure	QWK	+0.385	[0.341, 0.426]	[0.302, 0.460]	+0.592	[0.553, 0.627]	[0.517, 0.657]	+0.200	[0.145, 0.255]	[0.093, 0.304]
korean	Human	ρ	+0.044	[0.015, 0.073]	[-0.013, 0.104]	+0.053	[0.021, 0.084]	[-0.008, 0.116]	+0.064	[0.036, 0.093]	[0.010, 0.124]
korean	Human	QWK	-0.032	[-0.071, 0.008]	[-0.108, 0.043]	-0.239	[-0.276, -0.202]	[-0.308, -0.169]	-0.249	[-0.283, -0.214]	[-0.313, -0.183]
korean	Azure	ρ	+0.093	[0.052, 0.137]	[0.017, 0.182]	+0.227	[0.163, 0.296]	[0.110, 0.365]	+0.135	[0.094, 0.181]	[0.065, 0.231]
korean	Azure	QWK	+0.511	[0.470, 0.551]	[0.433, 0.585]	+0.422	[0.385, 0.457]	[0.351, 0.488]	+0.124	[0.082, 0.164]	[0.042, 0.199]
aesop	Human	ρ	+0.053	[0.023, 0.083]	[-0.005, 0.115]	+0.052	[0.019, 0.085]	[-0.013, 0.117]	+0.051	[0.022, 0.080]	[-0.006, 0.110]
aesop	Human	QWK	-0.465	[-0.493, -0.434]	[-0.518, -0.403]	-0.424	[-0.455, -0.391]	[-0.483, -0.361]	-0.459	[-0.488, -0.429]	[-0.513, -0.400]
aesop	Azure	ρ	+0.102	[0.063, 0.145]	[0.031, 0.189]	+0.226	[0.166, 0.291]	[0.117, 0.359]	+0.122	[0.080, 0.168]	[0.049, 0.219]
aesop	Azure	QWK	+0.079	[0.052, 0.103]	[0.026, 0.124]	+0.237	[0.209, 0.263]	[0.185, 0.287]	-0.087	[-0.124, -0.050]	[-0.159, -0.018]
koreanocean	Human	ρ	+0.040	[0.008, 0.071]	[-0.024, 0.101]	+0.032	[-0.002, 0.064]	[-0.035, 0.096]	+0.033	[0.002, 0.062]	[-0.027, 0.091]
koreanocean	Human	QWK	+0.115	[0.080, 0.151]	[0.048, 0.185]	+0.005	[-0.033, 0.042]	[-0.067, 0.077]	-0.067	[-0.102, -0.033]	[-0.134, -0.002]
koreanocean	Azure	ρ	+0.089	[0.047, 0.134]	[0.010, 0.180]	+0.206	[0.141, 0.276]	[0.086, 0.346]	+0.103	[0.059, 0.153]	[0.026, 0.206]
koreanocean	Azure	QWK	+0.659	[0.624, 0.690]	[0.591, 0.717]	+0.666	[0.627, 0.701]	[0.590, 0.731]	+0.305	[0.258, 0.351]	[0.214, 0.392]
aesopocean	Human	ρ	+0.062	[0.032, 0.092]	[0.006, 0.124]	+0.056	[0.024, 0.087]	[-0.004, 0.119]	+0.054	[0.026, 0.081]	[-0.000, 0.109]
aesopocean	Human	QWK	-0.097	[-0.131, -0.063]	[-0.160, -0.031]	-0.127	[-0.161, -0.093]	[-0.190, -0.062]	-0.261	[-0.292, -0.230]	[-0.320, -0.201]
aesopocean	Azure	ρ	+0.111	[0.070, 0.156]	[0.036, 0.203]	+0.230	[0.168, 0.297]	[0.118, 0.365]	+0.124	[0.079, 0.175]	[0.045, 0.231]
aesopocean	Azure	QWK	+0.446	[0.411, 0.480]	[0.379, 0.509]	+0.534	[0.500, 0.565]	[0.468, 0.592]	+0.111	[0.068, 0.152]	[0.029, 0.188]
koreanaesop	Human	ρ	+0.058	[0.029, 0.088]	[0.002, 0.119]	+0.071	[0.042, 0.101]	[0.016, 0.132]	+0.076	[0.049, 0.104]	[0.025, 0.133]
koreanaesop	Human	QWK	-0.232	[-0.265, -0.198]	[-0.294, -0.167]	-0.271	[-0.304, -0.237]	[-0.334, -0.205]	-0.308	[-0.340, -0.277]	[-0.368, -0.247]
koreanaesop	Azure	ρ	+0.107	[0.067, 0.151]	[0.033, 0.197]	+0.245	[0.183, 0.313]	[0.133, 0.382]	+0.147	[0.105, 0.194]	[0.075, 0.246]
koreanaesop	Azure	QWK	+0.312	[0.276, 0.345]	[0.245, 0.376]	+0.390	[0.356, 0.422]	[0.326, 0.451]	+0.064	[0.025, 0.102]	[-0.012, 0.135]
aesopkoreanocean	Human	ρ	+0.048	[0.018, 0.078]	[-0.010, 0.109]	+0.065	[0.034, 0.096]	[0.007, 0.127]	+0.065	[0.038, 0.092]	[0.012, 0.121]
aesopkoreanocean	Human	QWK	-0.129	[-0.163, -0.094]	[-0.193, -0.062]	-0.148	[-0.183, -0.113]	[-0.214, -0.081]	-0.224	[-0.256, -0.192]	[-0.285, -0.163]
aesopkoreanocean	Azure	ρ	+0.097	[0.056, 0.141]	[0.020, 0.187]	+0.239	[0.176, 0.308]	[0.125, 0.377]	+0.135	[0.093, 0.184]	[0.061, 0.237]
aesopkoreanocean	Azure	QWK	+0.415	[0.377, 0.450]	[0.343, 0.482]	+0.513	[0.477, 0.546]	[0.443, 0.576]	+0.148	[0.107, 0.188]	[0.067, 0.224]

Note. Positive differences favor the system; Human denotes human–other-rater consensus. Adjusted intervals cover 258 contrasts. Decisions use unrounded endpoints, not displayed zeros. QWK uses clipping and half-up rounding; Spearman retains raw scores.

B.1 Score distributions

The ridge distributions below use the Spearman-selected (S) condition; Azure is independent of selection rule. Means, sample standard deviations, bias, and MAE use unrounded predictions on the 0–100 scale against the three-rater mean; no recalibration is fitted.

Table 3: Score distributions for the S condition and Azure.

System	Accuracy				Fluency				Prosody			
	Mean	SD	Bias	MAE	Mean	SD	Bias	MAE	Mean	SD	Bias	MAE
azure	89.29	7.50	23.63	23.63	94.18	5.87	27.88	27.88	77.13	6.15	10.57	11.04
speechocean	73.57	6.14	7.90	8.96	64.67	7.37	-1.63	5.26	60.79	6.81	-5.76	6.70
korean	58.57	12.39	-7.10	7.66	53.31	10.75	-12.99	12.99	53.31	10.75	-13.25	13.25
aesop	38.49	13.31	-27.17	27.17	44.43	10.97	-21.87	21.87	44.14	10.62	-22.42	22.42
koreanocean	64.85	9.77	-0.81	4.38	57.68	8.45	-8.62	8.74	56.89	8.32	-9.66	9.72
aesopocean	55.27	11.66	-10.40	10.51	56.63	10.99	-9.67	9.79	55.25	9.89	-11.31	11.33
koreanaesop	50.09	14.23	-15.58	15.59	51.09	12.44	-15.21	15.21	50.26	12.36	-16.29	16.29
aesopkoreanocean	54.37	12.52	-11.30	11.38	54.18	10.50	-12.12	12.13	53.05	10.42	-13.50	13.50

C Public-development comparisons

The matched seven-family experiment uses fixed representation combinations and Spearman-based penalty selection (S) over included-corpus development sets. All entries are development Spearman correlations, not final-test results. SpeechOcean uses the matched fitting procedure, not the original historical deployment. Korean fluency and prosody share a source target. SO denotes SpeechOcean ($n = 500$), Korean has $n = 636$, and J-AESOP has $n = 36$ development recordings; the small J-AESOP sample limits precision.

Table 4: Public-development Spearman correlations for the S condition.

System	Accuracy			Fluency			Prosody		
	SO	Korean	J-AESOP	SO	Korean	J-AESOP	SO	Korean	J-AESOP
speechocean	0.816	0.699	0.967	0.886	0.668	0.856	0.890	0.653	0.929
korean	0.735	0.759	0.969	0.837	0.778	0.919	0.829	0.778	0.938
aesop	0.659	0.673	0.987	0.682	0.586	0.966	0.685	0.544	0.975
koreanocean	0.808	0.755	0.964	0.876	0.770	0.924	0.876	0.771	0.949
aesopocean	0.807	0.715	0.970	0.881	0.728	0.952	0.878	0.747	0.971
koreanaesop	0.683	0.755	0.976	0.834	0.779	0.959	0.830	0.777	0.973
aesopkoreanocean	0.804	0.748	0.971	0.875	0.769	0.953	0.872	0.770	0.974

D Korean supervision and representation sensitivity

D.1 Korean scoring and target mapping

Han, Chung, and Kim (2024; main-paper reference) define two 1–5 ratings: pronunciation accuracy and prosodic fluency. Accuracy addresses segmental clarity and errors; prosodic fluency combines stress, rhythm, intonation, speed, and pausing. The stored fields are `articulationScore` and `prosodyScore`. The second field supplies both fluency and prosody targets in the present implementation. Mapping $2.5(y-1)$ produces 0, 2.5, 5, 7.5, and 10 from the five integer levels; mapping does not equate the rubric with SpeechOcean or create an independent fluency annotation. Korean-only heads use the same features and targets for both constructs, yielding identical predictions in the evaluated configuration.

Source	Mapped	Accuracy (paraphrased)	Prosodic fluency (paraphrased)
1	0	Extensive segmental difficulty; understanding is hard.	Strongly disrupted prosody, very slow delivery, excessive pauses.
2	2.5	Frequent errors substantially burden understanding.	Awkward prosody and slow delivery with prolonged, poorly placed pauses.
3	5	Recurring errors interfere with understanding.	Uneven prosody and rate, with frequent or misplaced breaks.
4	7.5	Minor errors with little intelligibility impact.	Mostly steady delivery with mild prosodic or pausing irregularities.
5	10	Clear segments without noticeable errors.	Natural prosody, rate, and phrase boundaries.

D.2 Ablation protocol

The protocol was recorded before fitting on October 18, 2026 (repository: `results/ablation-mac-v1/PROTOCOL.md`). The completed public-development sweep crossed seven training families, three constructs, and all 15 nonempty subsets of four feature blocks, giving 315 configurations and 945 corpus-level comparisons. X denotes XEUS, R XLS-R, P PhoneticXeus embeddings, and I the 17 phone diagnostics; I still requires PhoneticXeus inference, so omitting P does not necessarily avoid that encoder. Existing speaker-disjoint partitions, corpus-balanced weights, training-only standardization, and block gains were preserved. Each configuration selected a penalty from $\{0.001, 0.01, 0.1, 1, 10, 100\}$ using equal-weight mean Spearman across all three development corpora, breaking ties by lower macro MAE and then larger penalty. This differs from the earlier included-corpus tuning; no corpus in these tables is an untouched cross-corpus test. Private recordings and ratings were excluded from fitting and selection.

Each candidate was compared with the fixed-feature baseline refitted under the same sweep protocol (X+P+I for accuracy, X+R otherwise). Candidates within 0.01 of the numerical best were shortlisted. Promotion required macro Spearman gain at least 0.01, no corpus correlation loss exceeding 0.02, no corpus MAE increase exceeding 0.10 on the 0–10 scale, and a positive lower 95% paired speaker-bootstrap bound (2,000 draws, seed 42). Fewer blocks were preferred where benefits were inconclusive. These screening gates do not constitute correction for all search comparisons. Deployed heads were not automatically replaced.

The decision table distinguishes numerical maxima from candidates passing the complete gate. Two AESOP-only alternatives passed: X+P for fluency and X+P+I for prosody. Their macro gains over the refitted AESOP baselines were 0.0341 (95% interval [0.0118, 0.0619]) for fluency and 0.0389 ([0.0161, 0.0638]) for prosody. Both exceed the 0.01 within-family threshold. Neither became the overall construct selection, and neither was substituted into the external seven-family evaluation. The overall retained choices were koreanocean X+P+I accuracy and aesopkoreanocean X+R fluency/prosody. Thus retaining the overall configurations does not mean that no individual family improved. XEUS-only macro correlations for the selected families were 0.8407 for koreanocean accuracy and 0.8644/0.8680 for aesopkoreanocean fluency/prosody, compared with 0.8424 and 0.8660/0.8721 for the retained combinations. These small development differences do not establish that all additional blocks are necessary; no XEUS-only private evaluation or matched resource benchmark is reported. The retained choices reflect the recorded development rule, not independently established representation optimality. Matrices report all 315 macro correlations; the accompanying CSV preserves all 945 corpus-specific correlations and MAEs.

Family	Construct	Numerical best	Best ρ	Retained/gated	Promotion
speechocean	accuracy	X+P+I	0.8276	X+P+I	No
speechocean	fluency	X+P+I	0.8655	X+R	No
speechocean	prosody	X+I	0.8640	X+R	No
korean	accuracy	X+R+I	0.8322	X+P+I	No
korean	fluency	X+I	0.8544	X+R	No
korean	prosody	X	0.8560	X+R	No
aesop	accuracy	X+P+I	0.7813	X+P+I	No
aesop	fluency	X+P	0.7961	X+P	Yes
aesop	prosody	X+P+I	0.7893	X+P+I	Yes
koreanocean	accuracy	X+P+I	0.8424	X+P+I	No
koreanocean	fluency	X+P	0.8685	X+R	No
koreanocean	prosody	X+P+I	0.8691	X+R	No
aesopocean	accuracy	X+P+I	0.8309	X+P+I	No
aesopocean	fluency	X+R+P+I	0.8654	X+R	No
aesopocean	prosody	X+R+P+I	0.8653	X+R	No
koreanaesop	accuracy	X+R	0.8337	X+P+I	No
koreanaesop	fluency	X+R+P	0.8600	X+R	No
koreanaesop	prosody	X+R	0.8599	X+R	No
aesopkoreanocean	accuracy	X+R+P+I	0.8425	X+P+I	No
aesopkoreanocean	fluency	X+R+P+I	0.8710	X+R	No
aesopkoreanocean	prosody	X+R	0.8721	X+R	No

D.3 Representation matrices

Columns S, K, A, SK, SA, KA, and SKA denote SpeechOcean, Korean, J-AESOP, and their respective combinations. Values are macro development Spearman correlations; X, R, P, and I denote XEUS, XLS-R, PhoneticXeus embeddings, and IPA diagnostics.

Table 6: Representation ablation by construct and training family.

Features	S	K	A	SK	SA	KA	SKA
Accuracy							
X	0.8208	0.8321	0.7655	0.8407	0.8226	0.8318	0.8336
R	0.6715	0.7460	0.3186	0.7646	0.7465	0.7766	0.7972
P	0.7860	0.7366	0.7292	0.8228	0.8065	0.7600	0.8273
I	0.6080	0.6742	0.6201	0.6961	0.3225	0.5164	0.4372
X+R	0.7957	0.8297	0.7414	0.8322	0.8257	0.8337	0.8361
X+P	0.8190	0.8030	0.7732	0.8350	0.8231	0.8159	0.8354
X+I	0.8253	0.8321	0.7522	0.8411	0.8249	0.8149	0.8362
R+P	0.7505	0.7690	0.6650	0.8192	0.8008	0.7871	0.8287
R+I	0.7290	0.7875	0.5625	0.7968	0.7765	0.7830	0.8085
P+I	0.7991	0.7750	0.7582	0.8308	0.8155	0.7171	0.8356
X+R+P	0.7962	0.8104	0.7453	0.8280	0.8235	0.8225	0.8350
X+R+I	0.7984	0.8322	0.7223	0.8363	0.8266	0.8183	0.8382
X+P+I	0.8276	0.8212	0.7813	0.8424	0.8309	0.8048	0.8409
R+P+I	0.7723	0.7932	0.7065	0.8260	0.8104	0.7740	0.8381
X+R+P+I	0.8048	0.8218	0.7565	0.8363	0.8304	0.8132	0.8425
Fluency							
X	0.8551	0.8543	0.7782	0.8630	0.8588	0.8545	0.8644
R	0.7312	0.7734	0.4373	0.8213	0.8000	0.8100	0.8319
P	0.8428	0.7912	0.7640	0.8592	0.8495	0.8127	0.8599
I	0.6288	0.6757	0.6577	0.6876	0.4060	0.5606	0.5002
X+R	0.8364	0.8444	0.7620	0.8567	0.8599	0.8574	0.8660
X+P	0.8613	0.8465	0.7961	0.8685	0.8645	0.8578	0.8702
X+I	0.8569	0.8544	0.7671	0.8618	0.8576	0.8538	0.8662
R+P	0.8178	0.8162	0.7153	0.8555	0.8498	0.8415	0.8653
R+I	0.7833	0.8076	0.6342	0.8407	0.8291	0.8200	0.8458
P+I	0.8469	0.8099	0.7748	0.8570	0.8511	0.8004	0.8593
X+R+P	0.8481	0.8458	0.7730	0.8609	0.8654	0.8600	0.8696
X+R+I	0.8443	0.8516	0.7400	0.8593	0.8598	0.8533	0.8652
X+P+I	0.8655	0.8508	0.7866	0.8666	0.8644	0.8498	0.8677
R+P+I	0.8340	0.8283	0.7268	0.8576	0.8515	0.8333	0.8653
X+R+P+I	0.8509	0.8481	0.7635	0.8639	0.8654	0.8556	0.8710
Prosody							
X	0.8631	0.8560	0.7817	0.8688	0.8599	0.8573	0.8680
R	0.7489	0.7740	0.3583	0.8172	0.7891	0.8178	0.8377
P	0.8313	0.7768	0.7520	0.8503	0.8416	0.7970	0.8526
I	0.6228	0.6834	0.6322	0.6960	0.4179	0.5379	0.5035
X+R	0.8410	0.8483	0.7504	0.8657	0.8653	0.8599	0.8721
X+P	0.8620	0.8384	0.7872	0.8664	0.8615	0.8508	0.8681
X+I	0.8640	0.8524	0.7627	0.8659	0.8609	0.8499	0.8693
R+P	0.8077	0.8070	0.6886	0.8517	0.8409	0.8296	0.8586
R+I	0.7789	0.8076	0.5893	0.8412	0.8229	0.8208	0.8498
P+I	0.8361	0.7979	0.7792	0.8527	0.8449	0.7793	0.8539
X+R+P	0.8444	0.8408	0.7602	0.8662	0.8647	0.8540	0.8707
X+R+I	0.8485	0.8512	0.7281	0.8663	0.8639	0.8495	0.8687
X+P+I	0.8635	0.8459	0.7893	0.8691	0.8649	0.8420	0.8690
R+P+I	0.8235	0.8210	0.7086	0.8522	0.8437	0.8214	0.8608
X+R+P+I	0.8485	0.8455	0.7554	0.8668	0.8653	0.8477	0.8715

E Execution environment

The same Windows/WSL machine used for the reported capacity test was inspected on October 19, 2026. The installation reports Ubuntu 26.04.1 LTS, kernel `6.18.54.1-microsoft-standard-WSL2`, Python 3.11.17, and

PyTorch `2.11.0+cu128` (CUDA build 12.8). Hardware comprises an Intel Core i7-11700K with 16 logical CPUs and an NVIDIA RTX 3080 with 10 GiB VRAM; the host has 32 GB RAM, with approximately 24 GiB allocated to WSL and 8 GiB swap. The reported inference capacity test used the CPU, not CUDA. These versions document the inspected installation; the archived timing and memory trace, rather than a new benchmark, supplies the reported performance measurements.

F External rater backgrounds and procedure

The original human-evaluation study reports three experienced university raters. The Korea-based rater had more than 20 years of teaching experience, including a decade of university pronunciation instruction and speaking assessment. The Japan-based rater had nearly 30 years of teaching experience and more than 10 years as a Cambridge English speaking assessor. The China-based rater had nearly 20 years of experience, including high-stakes speech rating and university speaking assessment in an international joint-degree program. All three held professorial positions, as confirmed by the study author.

Before evaluating the source recordings, the raters jointly assessed separate Stella recordings spanning proficiency levels and discussed discrepancies to establish shared criteria for segmental accuracy, fluency, and prosody. They then scored the 306 source recordings independently in randomized order, blinded to participant identity and automated scores. Ratings used a 0–100 scale in five-point increments, with communicative-effectiveness anchors labelled "High Intelligibility" above 85 and "Low Intelligibility" below 40. These are the documented general scale anchors, not a newly constructed analytic rubric for each dimension. The original study's authors served as its raters; their ratings were reused as the external reference for the present model evaluation. One recording was subsequently excluded for excessive duration, leaving the matched 305-recording cohort. Human calibration used separate recordings and did not fit or calibrate any automated scorer.